Krishiraj Santchurn

Civil & Structural Engineering · BIM + Computational · Commercially Aware

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PROFESSIONAL SUMMARY

Civil Engineering (Honours) student targeting graduate and entry-level roles in structural, civil and infrastructure engineering. Brings hands-on structural design experience from consultancy — reinforced-concrete design and AutoCAD documentation — alongside site and construction exposure on large-scale (~1,000-unit) housing developments, including with a national infrastructure ministry. CAPM-certified and fluent in core engineering software (Revit/BIM, Civil 3D, SpaceGass), with a commercial grounding in accounting and a record of project coordination and stakeholder engagement — differentiators that complement a solid technical foundation.

TECHNICAL SKILLS

Structural & Civil: Structural design, reinforced-concrete design, RC calculations, steel design, structural analysis, road design, site supervision, technical documentation, hands-on drafting
Software & BIM: Revit, BIM, Autodesk Civil 3D, AutoCAD, SpaceGass, ArcGIS Pro, Dynamo, digital-twin data
AI & Data: AI-assisted design and documentation workflows, prompt engineering, data analysis, digital-twin data
Commercial & Project: Project management (CAPM), accounting (ACCA), Microsoft Excel, Microsoft Office Suite, cost estimation, project schedule, conflict management, teamwork, visualisation
Fabrication: 3D printing, laser cutting, rapid prototyping

EDUCATION

Bachelor of Civil Engineering (Honours), Minor in Environmental Engineering

Monash University, Melbourne, Australia | Jul 2023 – Jul 2027

- Awarded the Sir John Monash Fee Scholarship and the Engineering International High Achievers Scholarship.

Advanced Diploma in Accounting and Business

Association of Chartered Certified Accountants (ACCA), United Kingdom | Jan 2022 – Dec 2024

EXPERIENCE

Structural Intern — Lux Consult

Mauritius | Dec 2025 – Jan 2026

- Produced reinforced-concrete design calculations (Excel) and full AutoCAD documentation for residential structures — villas, extensions and renovations — worked to senior-engineer review.
- Delivered 10+ technical drawings across 5+ structural design projects, improving documentation consistency and design turnaround.
- Applied structural design principles to real member sizing and detailing, translating engineering intent into buildable drawings.

Engineering Intern — Ministry of National Infrastructure

Mauritius | Dec 2024 – Jan 2025

- Supported site engineers on a large-scale (~1,000-unit) government housing development, tracking construction progress and assisting with quality and safety monitoring.
- Coordinated between engineers and contractors to keep concurrent works aligned on program and specification, strengthening site-delivery understanding.

Site Engineer Intern — Tayelamay Group

Mauritius | Nov 2023 – Feb 2024

- Assisted site engineers on a large (~1,000-unit) housing development, gaining direct exposure to construction sequencing, modular building methods and site coordination.
- Supported daily site activities, quality and safety checks, and progress reporting in an active construction environment.

Prototyping Division Sub-Lead — Monash Sustainable Buildings

Melbourne, Australia | Jul 2023 – Mar 2026

- Led the prototyping division for student-led sustainable building projects, producing components with CAD, 3D printing and laser cutting.
- Trained and onboarded new members and coordinated prototype design and fabrication across the team.

PROJECTS

Aboriginal Gathering Place — Structural Engineer (Final-Year Capstone)

2026 | Revit, SpaceGass, RC & Steel Design

- Leading the structural discipline on a community gathering place — developed and compared a reinforced-concrete scheme and a long-span structural-steel scheme for a fully column-free Multi-Purpose Room.
- Recommended the steel scheme for the clear span and faster construction, and am progressing it into a coordinated Revit structural model with member setout carried into a SpaceGass analytical model.

BIM Modelling in Revit — Self-Directed

2026 | Revit, BIM, Revit Structure

- Modelled a multi-unit residential apartment building end to end — unit layouts, circulation and a car park — producing plans, an elevation and a cutaway 3D view from the model.
- Extended it into a structure with a steel column-and-beam frame and a column-grid structural plan, plus supporting exercises on a two-storey house and a three-level frame with its analytical model.

Digital Twin Fire-Alarm Verification — Research Project

2026 | Digital Twin, Data Analysis

- Investigating how digital-twin data can reduce false fire alarms in institutional buildings, improving reliability and cutting unnecessary call-outs.
- Analysing building and sensor data to better distinguish genuine alarms from false triggers.

Rafflesia Vision Valley — Arterial Link Road Design

2025 | Autodesk Civil 3D, Austroads

- Planned and designed two Austroads-compliant arterial link roads for a logistics development, from alignment options to a final planning map produced in Civil 3D.
- Applied road geometry, drainage and setout standards to deliver a coordinated, standards-compliant design package.

LEADERSHIP & ACTIVITIES

Founder & President — Monash Mauritian Society

Nov 2024 – Present

- Founded a cultural student society and secured AUD $4,000 in sponsorship through stakeholder engagement and end-to-end event planning.
- Grew it into a Melbourne-wide Mauritian student community spanning six universities.

Founding Group Leader — AWS Student Builder Group, Monash

Sep 2026 – Present

- Selected to establish Monash's AWS cloud-computing student community from the ground up — structure, events and membership.

Committee Member — ACES, Monash Association of Civil Engineering Students

Apr 2024 – Oct 2026

- Supported student engagement through academic and industry initiatives, including the annual Industry Night.
- Helped connect students with industry through events such as the ICE/SRLA night and SMEC “Consultant for a Day”.

Campus & Student Ambassador — Monash University

May 2024 – Present

- Represented the Faculty of Engineering and the university across recruitment, outreach and student engagement.

CERTIFICATIONS

- CAPM® – Certified Associate in Project Management, Project Management Institute (2026)
- buildingSMART Professional Certification – Entry, openBIM / IFC (2026)
- SEED-AI – Skills for Education, Employment and Digital AI, Deakin University (2026)
- CPCWHS1001 – Construction Induction (White Card)
- Taekwondo 1st Dan Black Belt – Kukkiwon (World Taekwondo Headquarters), Cert. No. 05752827 (2022)

INTERESTS

- Competitive martial arts: 14 years of Taekwondo (Kukkiwon 1st Dan Black Belt); currently train wrestling and Brazilian Jiu-Jitsu (BJJ) daily — building discipline, resilience and strong time management alongside full-time study.